10. Hash Table :

```c
#include <stdio.h>
# define MAX 20
int hashTable [MAX];
int m;
void insert (int key){
    int index = key % m;
    if (hashTable [index] == -1) {
        hashTable [index] = key;
    }
    else{
        int i = 1;
        while (hashTable [(index+i) % m] != -1)
            i++;
        hashTable [(index+i) % m] = key;
    }
}
void display (){
    printf("\n Hash Table : \n");
    for (int i=0; i<m; i++){
        if (hashTable[i] != -1)
            printf(" Address %d: %d \n", i, hashTable[i]);
```

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```
        else
            printf("Address %d : Empty\n", i);
    }
}
int main() {
    int n, key;
    printf(" Enter size of hash table (m): ");
    scanf("%d", &m);
    printf(" Enter no. of employee records: ");
    scanf("%d", &n);
    for (int i=0; i<m; i++)
        hashTable[i] = -1;
    printf(" Enter %d employee keys (4-digit): \n", n);
    for (int i=0; i<n; i++) {
        scanf("%d", &key);
        insert(key);
    }
    display();
    return 0;
}
```

Output:

```
Enter size of hash table (m): 7
Enter no. of employee records: 4
Enter 4 employee keys (4-digit):
1234
2345
3456
4567

Hash Table:
Address 0: 2345
Address 1: Empty
Address 2 : 1234
Address 3 : 4567
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Address 4 : Empty
Address 5 : 3456
Address 6 : Empty.